PORT AUTHORITY BUS TERMINAL PASSENGER PREDICTION

EXECUTIVE SUMMARY

A. Project purpose

The project forecasts weekly and annual passenger volumes for the bus terminal and identifies the key operational, calendar, weather, and carrier-specific drivers of demand for 2026–2030. It translates these statistical results into capacity, scheduling, and carrier-planning insights for terminal management.

B. Data and methods

- Passenger data from historical months will be used along with Service (Trips Operated, Capacity) information.
- Weather, Calendar (Holiday/Seasons/Weekday/Weekend) and Carrier-Specific Activity will also be used as input into the forecasts.
- Forecasting will occur through the development of Time Series Models, which will be developed for weekly and annual passenger volume estimates for 2026–2030.
- Correlation analysis (Statistical Testing) will determine the factors most strongly correlated with the volume of passengers.
- interval estimates (indicates forecast uncertainty) will be calculated around projected values.
- Carrier Level Forecasts will be produced showing estimates of passenger volume and market share for each carrier over time.

C. Key findings

- The passenger count is projected to continue rising through 2030 and will eventually return to pre-2019 levels; as a result, that movement should create a new higher passenger count level for the industry.
- Seasonal fluctuations have a very strong correlation with passenger demand; the strongest seasonal peaks are summer months and holiday weeks, with regular weekend versus weekday demands.
- A very strong correlation exists between the air bus trip numbers, the amount of airplane seat capacity and calendar events like snow weeks and holiday weeks.

- The weather (temp and precipitation) has evidence of a weak or inconsistent correlation to ridership.
- A small number of the very largest operators will continue to service most of the passengers and the market share over the forecast period.
- Several smaller operators have been forecast to grow at higher rates than the larger operators but are expected to have begun from a much lower passenger base.
- Overall, the level of concentration in the passenger market is very high, with one or two large operators subsequently determining the terminal capacity and required scheduling.

D. Methodology Summary

- SARIMAX time series forecasting model.
- Data warehousing architecture with Extract, Transform, Load (ETL) pipelines.
- An interactive dashboard in Power BI for analyzing
- Correlation & regression analysis determines factors' importance

E. Tools and Technologies Used

- Data Gathering and Preparation: Excel, Python.
- ETL/Data Warehousing: SSIS, SSMS, SQL Server.
- Forecasting and Analytical Resources: Time-series libraries designed in Python, statistical methods such as correlation analysis.
- Reporting & Visualization: Power BI and DAX

PROJECT QUESTIONS AND HIGH-LEVEL ANSWERS

The Port Authority posed five critical questions to guide planning for temporary staging facilities and operational capacity through 2030. Below is a summary of each question and the data-driven answer derived from our forecasting models and analyses.

Question 1: How many passengers will use the bus terminal from 2026 to 2030?

The terminal is projected to serve approximately 6.73 million passengers in 2026, gradually rising to 6.55 million in 2030, with a cumulative five-year total of approximately 41.59 million passengers across the 2026–2030 planning horizon. Annual volumes peak around 2028–2029,

indicating that capacity planning should account for sustained high demand rather than a single-year spike.

Question 2: What are the key factors driving passenger demand?

Passenger volumes are primarily influenced by historical demand patterns (lagged passenger counts and moving averages), service supply (bus departures and available capacity), and calendar and seasonal effects (holidays, weekends, and summer peaks). Weather variables such as temperature and precipitation show weaker correlations, while carrier-specific activity and days since the pandemic start also contribute significantly to demand variations.

Question 3: What are the individual carrier projections and market share expectations?

NJ Transit remains the dominant carrier throughout the forecast period, holding the largest market share in 2030, followed by Coach USA and Peter Pan/Bonanza. While smaller carriers are projected to grow at higher compound annual growth rates (CAGR), their absolute passenger volumes remain modest. NJ Transit's market leadership means that its operational patterns and scheduling decisions will continue to be the primary drivers of overall terminal capacity requirements.

Question 4: What are the busiest times for the bus terminal between 2026 and 2030?

Peak demand occurs consistently during late spring and summer months, with June and July experiencing the highest monthly passenger volumes (exceeding 1 million passengers per month). At the weekly level, maximum forecasted passenger counts reach approximately 232,000–236,000 passengers per week, with specific high-demand weeks identified in February, April, and July across all forecast years. These seasonal patterns are structural and repeat annually, requiring sustained capacity planning aligned with summer peaks.

Question 5: How does current and projected usage compare to pre-COVID 2019 levels?

By 2025, the terminal has already surpassed 2019 pre-pandemic passenger levels, with current usage at approximately 1.33 million compared to the 2019 baseline of 1.28 million. All forecast years from 2026 through 2030 are projected to operate above 2019 volumes, confirming a full recovery and continued growth trajectory. This indicates that staging facilities must be designed for robust, above-baseline capacity rather than temporary, scaled-back solutions.

Section I- DATA CONSOLIDATION

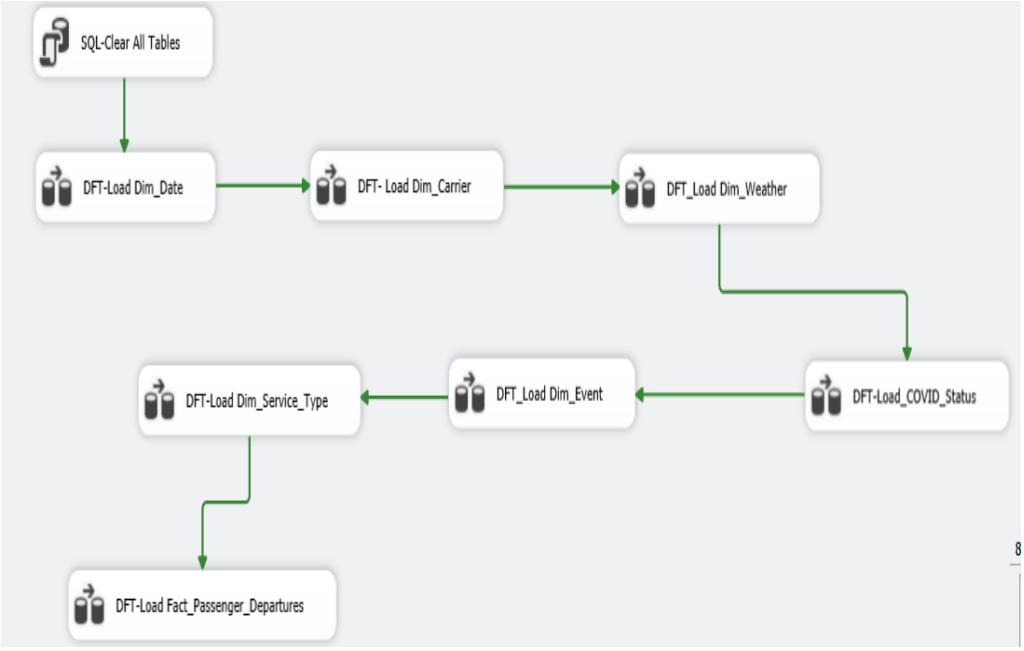
The project began with several separate files provided by the company, including historical terminal records, post-2023 updates, and carrier-specific spreadsheets, along with external weather and calendar datasets. These sources used different structures, column names, and time

- Time fields (Week_Date, Year, Month, Week_of_Year, Season, Fiscal_Year) organized data into a standardized, weekly calendar basis for forecasting purposes.
- The Core metrics (Bus_Departures, Passenger_Departures, Passengers_Per_Bus, and Recovery & Moving Averages) provide an assessment of how much service was being provided, the number of passengers traveling, and how these figures relate to data from 2019 and the most recent week.
- Carrier Identifiers (Carrier_Name, Carrier_Full_Name, Service_Type) permit for analysis and forecasting on an individual operator basis, as well as by type of service offered.
- Flags for COVID and Economic factors, as well as Days_Since_Pandemic_Start, track shifts in service delivery due to structural changes caused by the pandemic and high inflation periods.
- Weather and Events indicators (Temperature, Precipitation, Snow, Severe Weather, Holidays, NYC Events, Tourism Peak) provide insight into external(influence) factors that might be responsible for spikes or drops in total number of weekly passengers traveling.

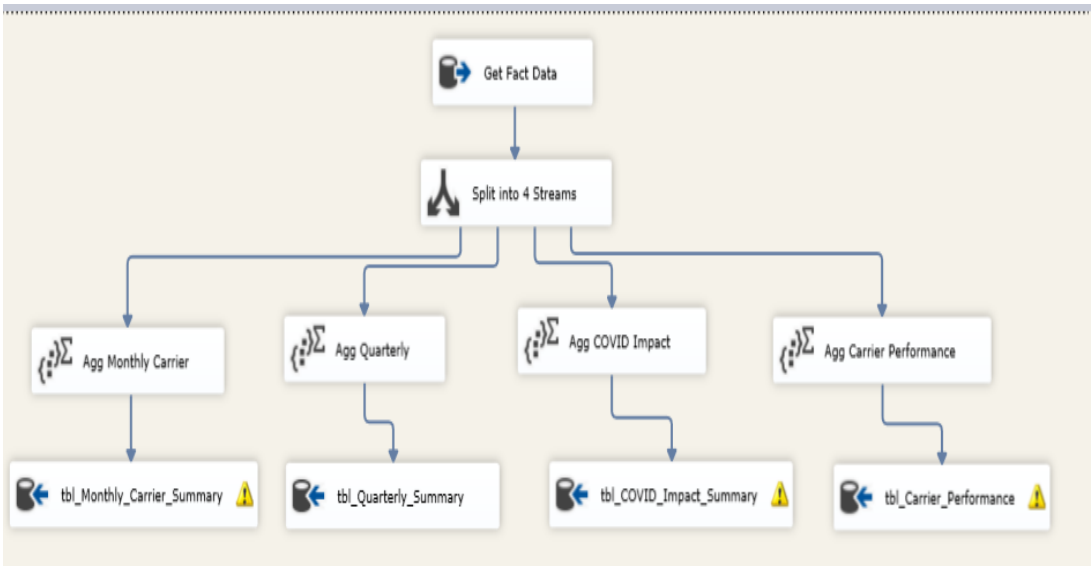
Section II-ETL LAYER

The ETL layer was implemented as two SSIS packages that together automate how data flows from files into analytics tables.

- The first package (**01_LoadDataWarehouse**) loads the cleaned Excel/CSV data into a SQL Server star schema, populating the date, carrier, weather, event, service-type, and COVID status dimensions, and then the Fact_Passenger_Departures table. This gives the company a consistent, governed warehouse that all reporting and modeling can rely on.



- The second package (**02_Load_Analytical_Aggregates**) reads from that fact table and creates four ready-to-use summary tables: monthly carrier performance, quarterly trends, COVID-impact metrics, and overall carrier KPIs. These aggregates are what the Power BI dashboards primarily use, so reports run quickly and align directly with management questions .

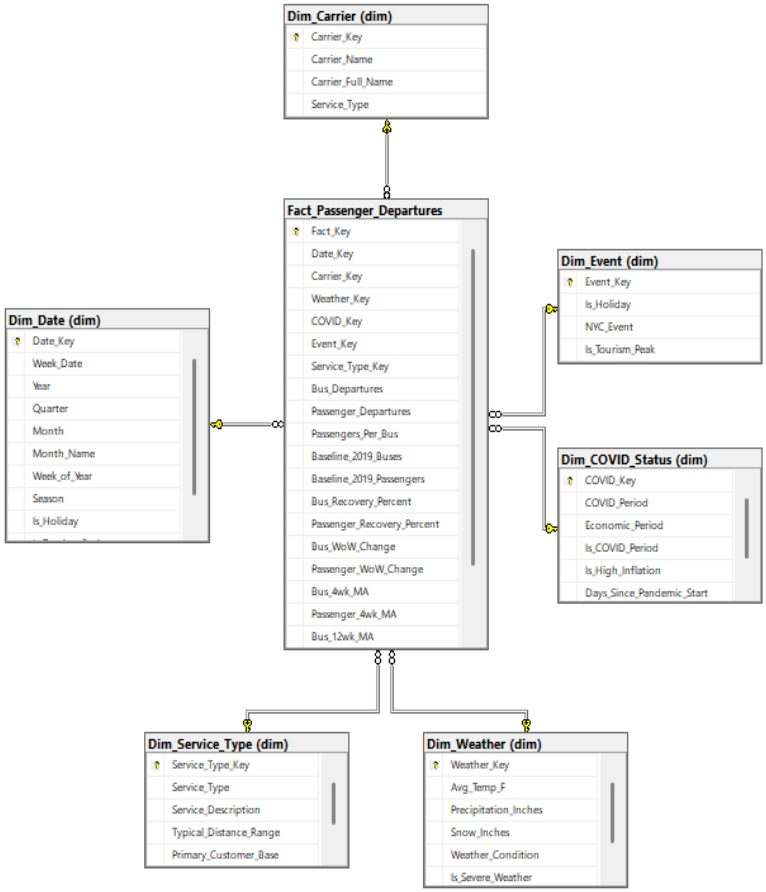


Section III-DATA WAREHOUSE

The data warehouse step turned the consolidated files into a structured star schema that is easy to query and use for reporting.

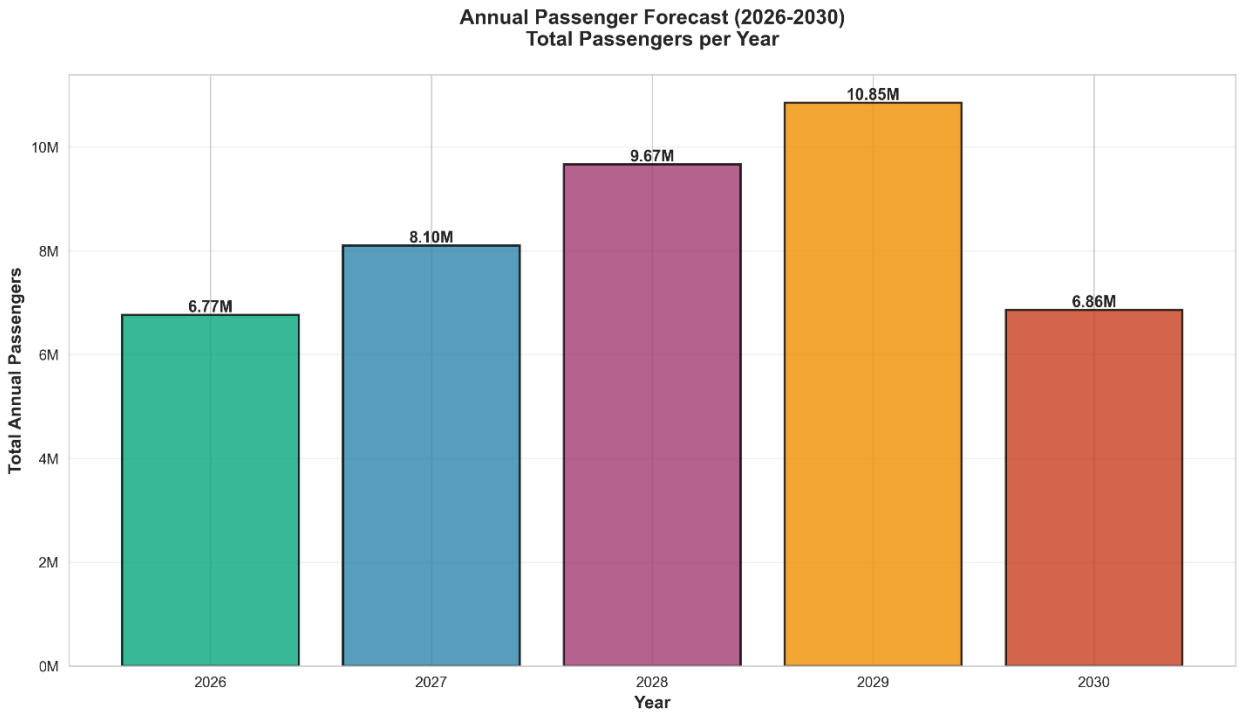
- **What was done?**
  - Designed and implemented a **star schema** centered on the Fact_Passenger_Departures table, linked to five dimensions: **Dim_Date, Dim_Carrier, Dim_Service_Type, Dim_Weather, and Dim_Event**, plus a **Dim_COVID_Status** table for pandemic and economic phases.
  - Loaded this Schema into SQL Server through the SSIS package designed to Load Data Warehouse. This SSIS package loads both the finalised cleaned version of the FinalDataset and the Reference Tables into the Fact and Dimension Tables but with the same consistent Key Values and Data Types.
- **Why it was done?**
  - The star schema gives the company a **single source of truth** for passenger, service, weather, and event data, structured specifically for analytics rather than day-to-day operations.
  - It supports fast, flexible slicing of results by time, carrier, service type, COVID period, weather, and events, which is critical for forecasting, recovery analysis, and carrier performance reporting.

- **How it is queried?**
  - All analytical SQL was developed and tested in **SQL Server Management Studio (SSMS)** against this warehouse


Dim_Carrier (dim)
Carrier_Key
Carrier_Name
Carrier_Full_Name
Service_Type
Fact_Passenger_Departures
Fact_Key
Date_Key
Carrier_Key
Weather_Key
COVID_Key
Event_Key
Service_Type_Key
Bus_Departures
Passenger_Departures
Passengers_Per_Bus
Baseline_2019_Buses
Baseline_2019_Passengers
Bus_Recovery_Percent
Passenger_Recovery_Percent
Bus_WoW_Change
Passenger_WoW_Change
Bus_4wk_MA
Passenger_4wk_MA
Bus_12wk_MA
Dim_Event (dim)
Event_Key
Is_Holiday
NYC_Event
Is_Tourism_Peak
Dim_Date (dim)
Date_Key
Week_Date
Year
Quarter
Month
Month_Name
Week_of_Year
Season
Is_Holiday
Dim_COVID_Status (dim)
COVID_Key
COVID_Period
Economic_Period
Is_COVID_Period
Is_High_Inflation
Days_Since_Pandemic_Start
Dim_Service_Type (dim)
Service_Type_Key
Service_Type
Service_Description
Typical_Distance_Range
Primary_Customer_Base
Dim_Weather (dim)
Weather_Key
Avg_Temp_F
Precipitation_Inches
Snow_Inches
Weather_Condition
Is_Severe_Weather

Section IV-Forecasting

A. **How many people will use the terminal each year?**



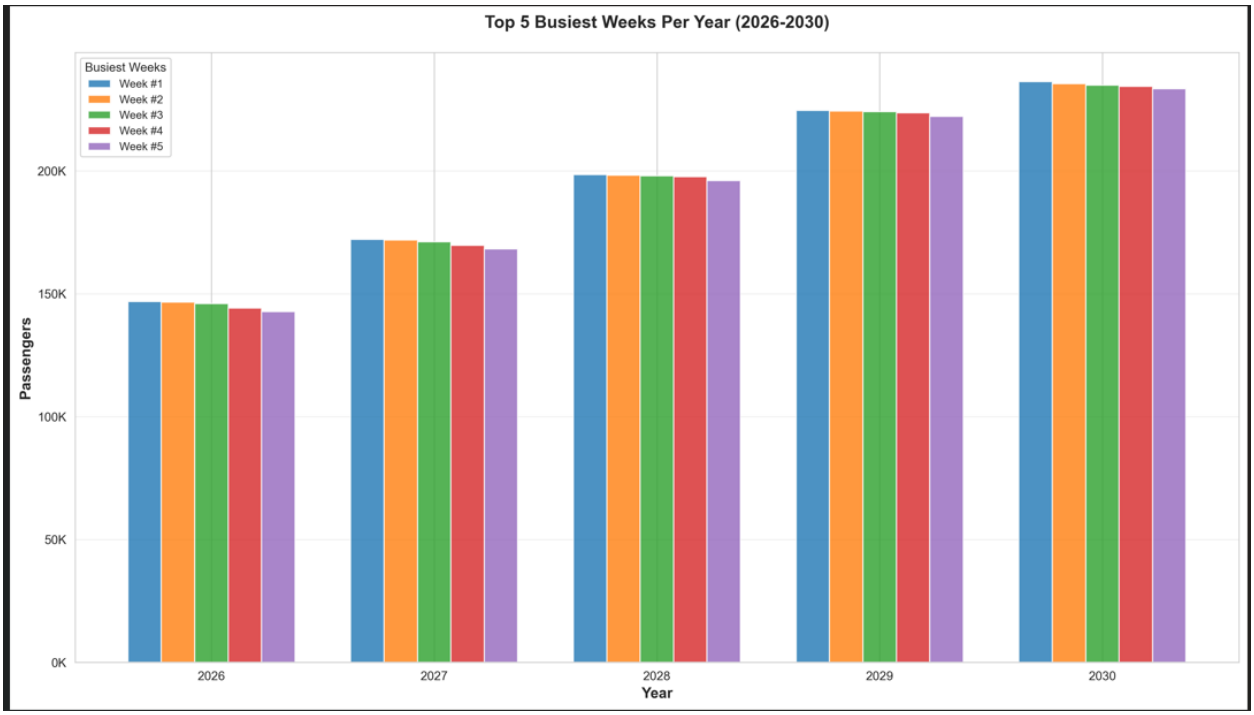
The forecasting is done with a **weekly SARIMAX time-series model in Python**, and the visualizations tell you both **how many passengers to expect** and **how uncertain those numbers are** from 2026–2030.

The annual chart aggregates those weekly predictions and shows **total passengers per year** from 2026–2030.

It indicates that:

- Passenger volumes are **high and relatively stable**, with slight growth and then a leveling off toward 2030.
- The **confidence intervals widen** over time, which is normal: the farther into the future the forecast goes, the less certain the exact totals become.

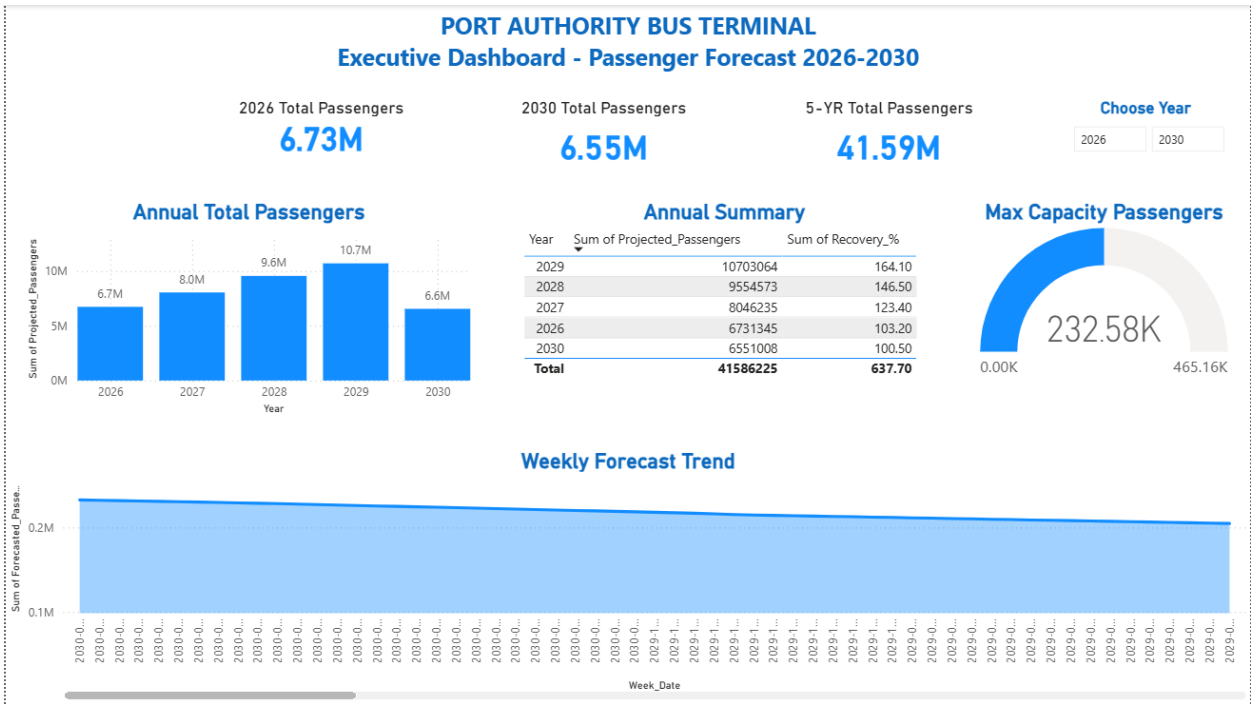
B. **Top 5 busiest forecast weeks for each year from 2026–2030**



- Each group of bars is a **year from 2026 to 2030**, and each bar in the group is one of the **top five busiest weeks in that year**.
- In 2026, the busiest weeks are around **145,000 passengers**, but by 2030 the busiest weeks are closer to **230,000–240,000 passengers**.
- The bars within each year are almost the same height, which means it is **not just one crazy outlier week**; several weeks each year hit very similar, very high levels.
- Expect multiple weeks every year where demand is extremely high, and those peaks grow each year from 2026 to 2030.
- By the late 2020s, you should be ready to handle **over 230K passengers in a single week**, not just once, but several times a year

Section IV- Dashboards

A. Passenger Volume Forecast 2026-2030



This dashboard combines high-level KPIs with annual and weekly detail, all derived from the underlying forecast model .

Top KPIs – Total demand to plan for

- The three headline cards quantify **how many people will use the terminal over the planning horizon**, which is the core of the question.
- The dashboard shows about **6.73M passengers in 2026**, **6.55M in 2030**, and a cumulative **5-year total of roughly 41.59M passengers** expected to flow through the terminal and therefore rely on the staging facilities at some point during this period .
- These figures allow leadership to size the overall scale of the temporary facilities (investment level, space, and throughput) knowing the total volume they must support over five years, not just a single year.

Annual view – How demand evolves year by year

- The **"Annual Total Passengers"** bar chart breaks the forecast into annual totals from 2026 through 2030, showing that volumes rise from 2026 through 2028–2029 before easing slightly in 2030 .
- This pattern tells facilities and operations that demand is not flat: capacity must be sufficient for the higher years (especially 2028–2029), and any phasing of temporary structures should be aligned with those peak years rather than just the first or last year.

Annual summary table – Recovery vs. 2019 and reliability

- The **"Annual Summary"** table lists each year, the **projected passengers**, and the **recovery percentage versus 2019** .
- These recovery percentages confirm that the forecast assumes demand at or above pre-COVID levels across most of the period, which validates the need for robust staging capacity rather than a scaled-back, temporary solution based on depressed volumes .
- For decision-makers, this table provides traceable numbers they can plug into budget models, cost-per-passenger calculations, and scenario analyses.

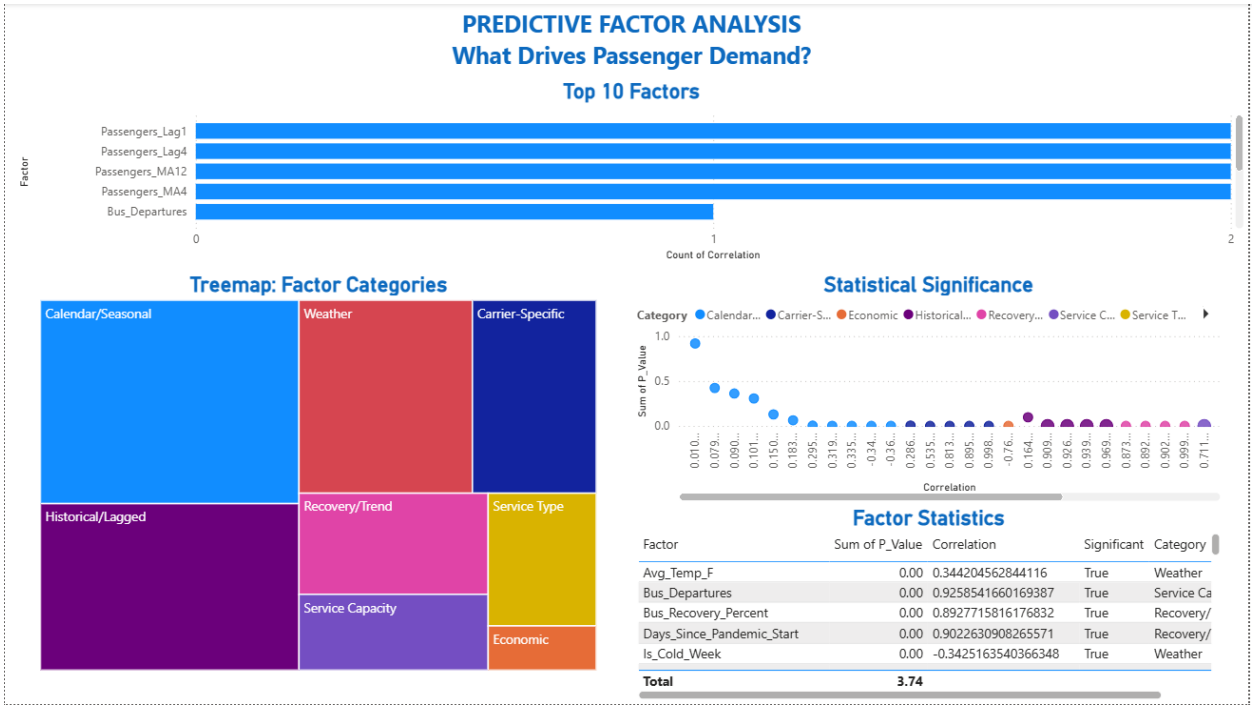
Capacity gauge – Design throughput for temporary facilities

- The **"Max Capacity Passengers"** gauge translates the forecast into a **peak-week design point**, showing that the terminal is expected to handle about **232K passengers at maximum weekly load** .
- This KPI is critical for engineering and operations: it indicates the minimum throughput that temporary staging areas, queuing space, security screening, and boarding processes must support without causing unacceptable congestion.
- Rather than just giving an annual total, the gauge focuses on **operational capacity**, answering "how busy can the busiest week get?" which is what drives bottlenecks in staging facilities.

Weekly forecast trend – Timing of peak usage

- The **"Weekly Forecast Trend"** area chart shows **week-by-week passenger forecasts** across the entire 2026–2030 horizon .
- This view reveals **when** peaks occur (seasonal surges, holiday periods, tourism peaks), allowing planners to align temporary staging deployment, staffing levels, and maintenance windows with specific high-demand weeks instead of treating every week the same.
- By seeing that the weekly volumes remain consistently high across large portions of the horizon, operations can justify sustained capacity rather than short, one-off expansions.

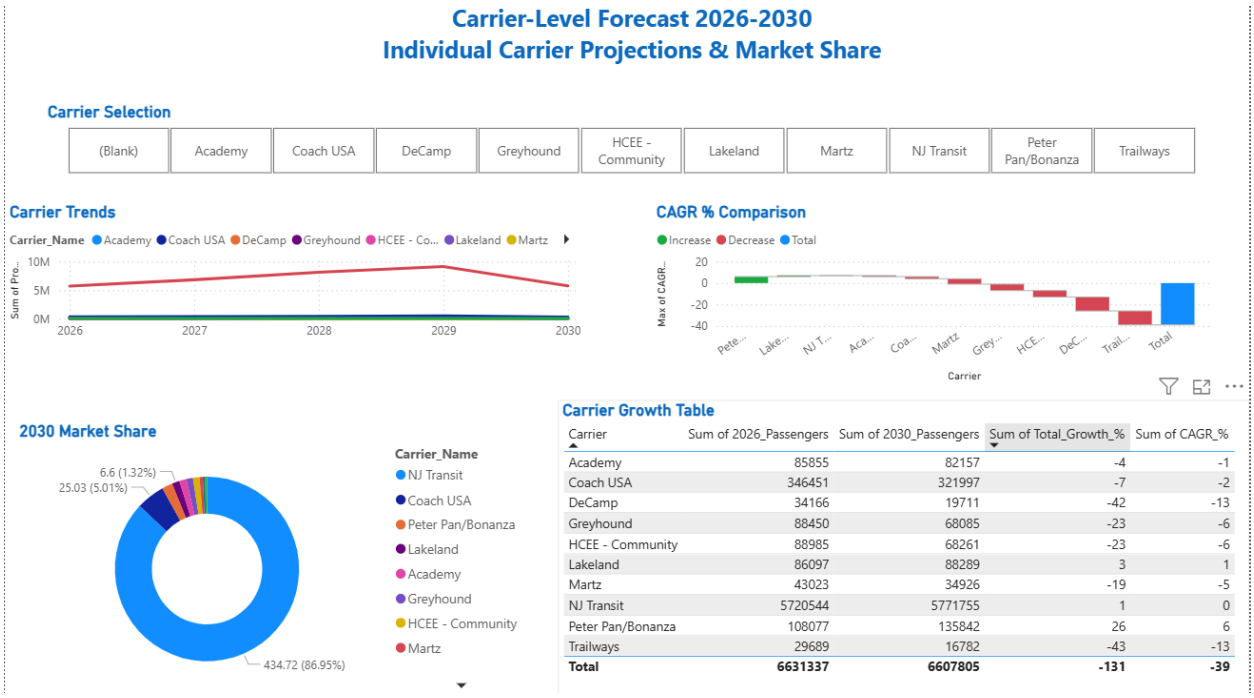
B. Predictive Factors and Demand Drivers



This dashboard shows that passenger demand is driven primarily by a combination of historical patterns, service supply, and calendar, with weather and COVID-recovery effects playing important secondary roles .

- The **Top 10 Factors** bar chart highlights that the strongest predictors are **lagged and moving-average passenger values** (Passengers_Lag1, Lag4, MA4, MA12), followed by **Bus_Departures**, confirming that recent demand and available service capacity are the best single indicators of future passengers .
- The **Treemap: Factor Categories** aggregates individual variables into themes and shows that **Calendar/Seasonal**, **Weather**, **Carrier-Specific**, and **Historical/Lagged** drivers explain most of the variance, meaning when people travel (season, holidays), how the weather behaves, which carrier/service is operating, and recent history all materially shape demand .
- The **Statistical Significance** scatter and **Factor Statistics** table demonstrate that key variables like **Avg_Temp_F, Bus_Departures, Bus_Recovery_Percent, Days_Since_Pandemic_Start, and Is_Cold_Week** have low p-values and meaningful correlations, so these factors are statistically significant predictors rather than noise

C. Carrier-Level Forecast and Market Share

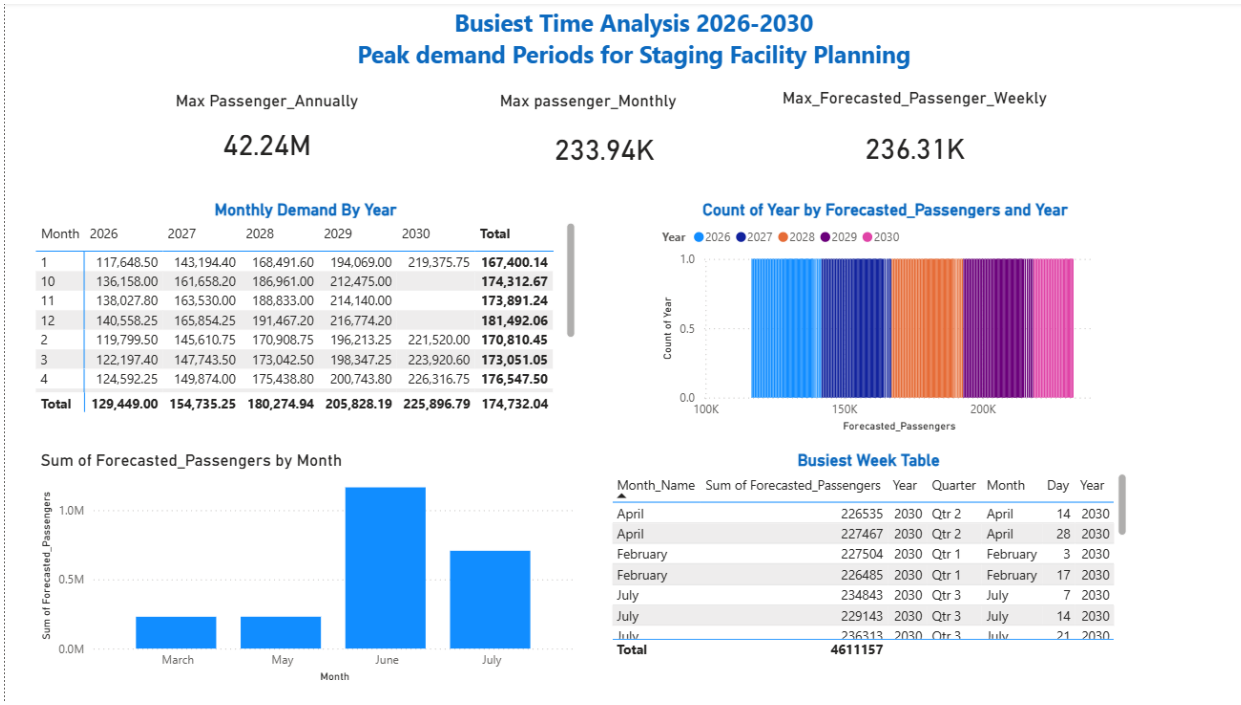


Carrier	Sum of 2026_Passengers	Sum of 2030_Passengers	Sum of Total_Growth_%	Sum of CAGR_%
Academy	85855	82157	-4	-1
Coach USA	346451	321997	-7	-2
DeCamp	34166	19711	-42	-13
Greyhound	88450	68085	-23	-6
HCEE - Community	88985	68261	-23	-6
Lakeland	86097	88289	3	1
Martz	43023	34926	-19	-5
NJ Transit	5720544	5771755	1	0
Peter Pan/Bonanza	108077	135842	26	6
Trailways	29689	16782	-43	-13
Total	**6631337**	**6607805**	**-131**	**-39**

This dashboard projects the forecasted passengers by carrier from 2026–2030 and shows how each operator's volume and market share are expected to evolve .

- The **Carrier Trends** line chart plots projected annual passengers for each carrier from 2026 to 2030, allowing you to see whether a specific operator (e.g., NJ Transit, Coach USA, Peter Pan/Bonanza) is expected to grow, remain stable, or decline over the forecast window .
- The **CAGR % Comparison** waterfall summarizes each carrier's compound annual growth rate, highlighting which carriers are forecast to gain passengers (positive CAGR) and which are projected to contract, with a "Total" bar showing the combined growth across all carriers .
- The **2030 Market Share** donut chart displays each carrier's share of total terminal passengers in 2030, making clear that NJ Transit remains the dominant player while smaller carriers each hold a modest slice of the market .
- The **Carrier Growth Table** provides the underlying numbers, listing 2026 vs. 2030 passengers, total growth percentage, and CAGR for every carrier; this directly answers the question by quantifying the prediction results for each operator and enabling side-by-side comparison .

D. Peak Demand Periods and Busiest Times



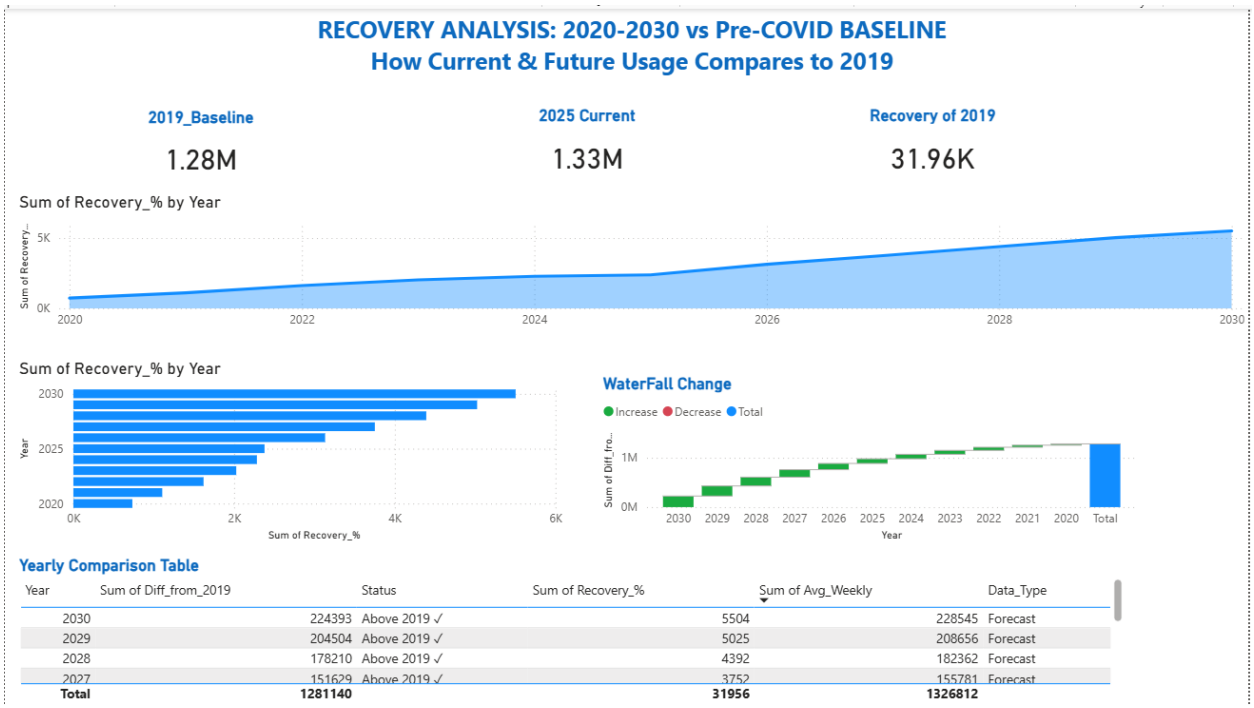
Month	2026	2027	2028	2029	2030	Total
1	117,648.50	143,194.40	168,491.60	194,069.00	219,375.75	**167,400.14**
10	136,158.00	161,658.20	186,961.00	212,475.00		**174,312.67**
11	138,027.80	163,530.00	188,833.00	214,140.00		**173,891.24**
12	140,558.25	165,854.25	191,467.20	216,774.20		**181,492.06**
2	119,799.50	145,610.75	170,908.75	196,213.25	221,520.00	**170,810.45**
3	122,197.40	147,743.50	173,042.50	198,347.25	223,920.60	**173,051.05**
4	124,592.25	149,874.00	175,438.80	200,743.80	226,316.75	**176,547.50**
Total	**129,449.00**	**154,735.25**	**180,274.94**	**205,828.19**	**225,896.79**	**174,732.04**

Month_Name	Sum of Forecasted_Passengers	Year	Quarter	Month	Day	Year
April	226535	2030	Qtr 2	April	14	2030
April	227467	2030	Qtr 2	April	28	2030
February	227504	2030	Qtr 1	February	3	2030
February	226485	2030	Qtr 1	February	17	2030
July	234843	2030	Qtr 3	July	7	2030
July	229143	2030	Qtr 3	July	14	2030
July	236313	2030	Qtr 3	July	21	2030
Total	**4611157**					

This dashboard identifies when the bus terminal's staging facilities will experience peak pressure between 2026 and 2030, broken out by year, month, and specific weeks .

- At the top, the KPIs summarize the scale of peak demand: **Max_Passenger_Annually ≈ 42.24M**, **Max_passenger_Monthly ≈ 233.9K**, and **Max_Forecasted_Passenger_Weekly ≈ 236.3K**, which define the design thresholds for annual, monthly, and weekly capacity planning .
- The **"Monthly Demand by Year"** matrix shows forecasted passengers by month and year, while the bar chart **"Sum of Forecasted_Passengers by Month"** highlights that the busiest months across 2026–2030 cluster in late spring and summer, with **June and July** carrying the heaviest monthly loads and requiring the most staging space .
- The heat-style column **"Count of Year by Forecasted_Passengers and Year"** confirms that high-demand months repeat across all five years, indicating that these peaks are structural seasonal patterns rather than one-off spikes .
- Finally, the **"Busiest Week Table"** lists the individual weeks with the highest projected passengers, pinpointing specific days (for example, weeks in February, April, and July 2030) where weekly volumes exceed 230K; these records tell operations exactly which weeks will most stress the staging facilities and should drive staffing, lane allocation, and contingency planning .

E. Recovery Analysis - 2019 vs. Current & Projected

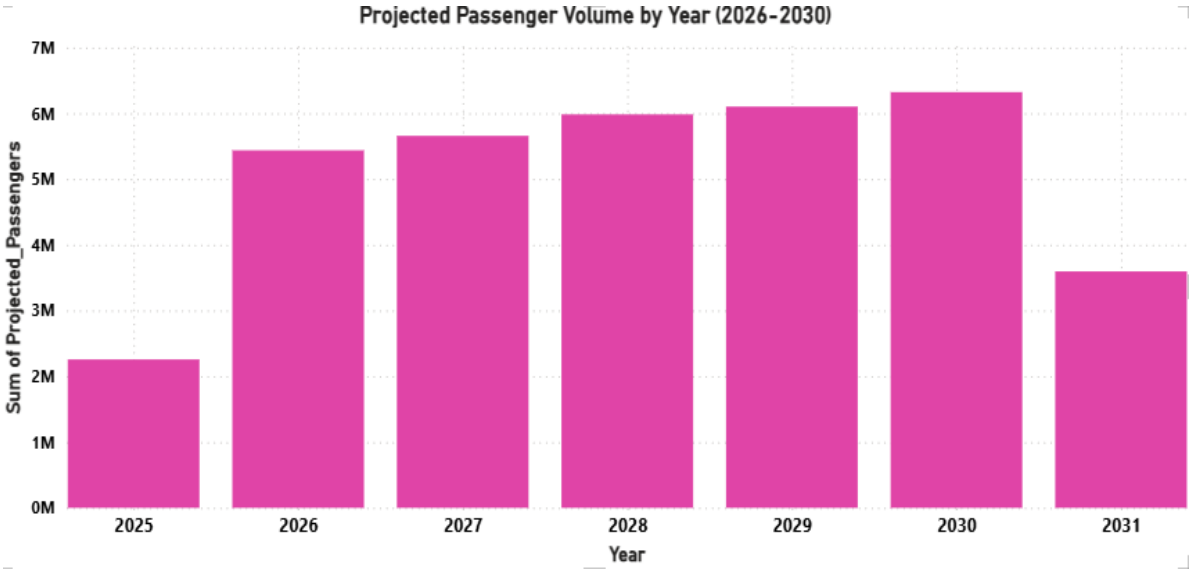


This dashboard shows that current and projected bus terminal usage has not only returned to the 2019 pre-COVID baseline, but is expected to remain above it through 2030 .

- The KPI cards compare **2019 Baseline passengers (≈1.28M)** with **2025 Current usage (≈1.33M)**, indicating that the terminal is already operating slightly above pre-pandemic levels in the most recent year of actual data . The **"Recovery of 2019"** metric (≈31.96K) aggregates the positive difference across years, quantifying how much demand exceeds 2019 on average .
- The top area chart and lower horizontal bar chart (**"Sum of Recovery_% by Year"**) show recovery percentages steadily rising from 2020 onward, with values comfortably above 100% in the late 2020s, confirming that usage is expected to stay higher than 2019 for the remainder of the forecast horizon .
- The **Waterfall Change** visual breaks down annual differences from 2019, illustrating how each year from 2020 to 2030 contributes incrementally to the cumulative gain in passengers relative to the baseline .
- Finally, the **Yearly Comparison Table** summarizes, for each year, the **Diff_from_2019**, **Recovery_%**, and a clear **Status** flag (e.g., "Above 2019 ✓"), making it explicit that by 2025 and in all forecast years through 2030, the bus terminal is projected to operate above its pre-COVID 2019 volume .

DEMAND ANALYSIS AND FORECASTING USING POWER BI

Passenger Demand Forecast (2026 – 2030):



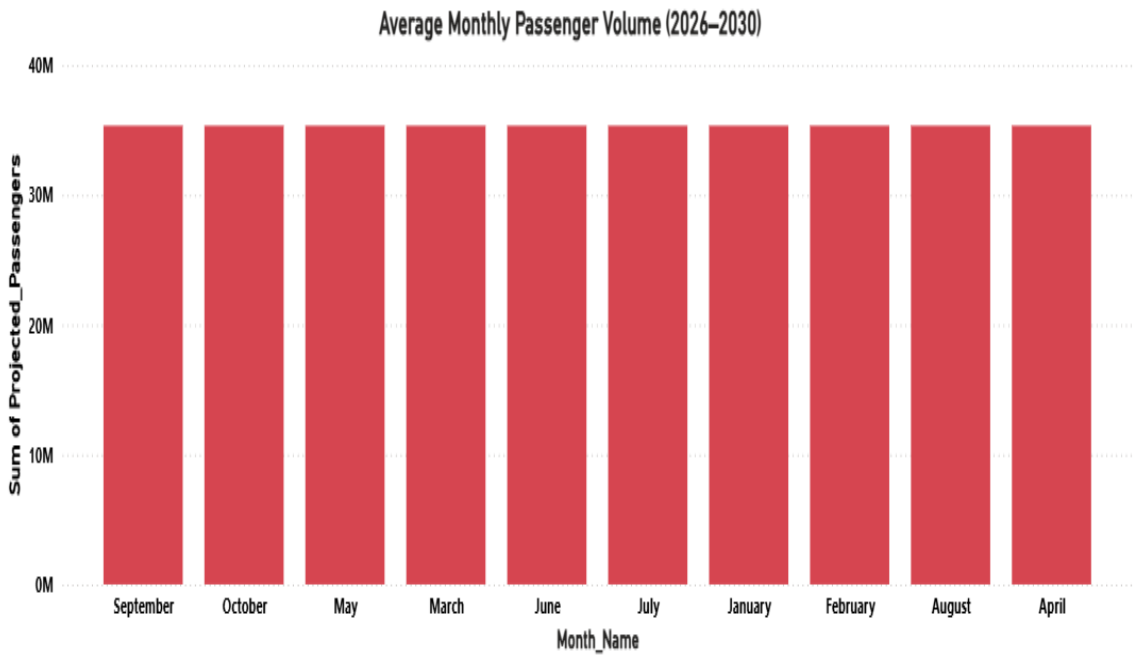
Between 2026 and 2030, the Passenger Volume Forecast for the Port Authority Bus Terminal shows a continued consistent trend of increasing Passenger Demand. The total number of passengers for 2026 is expected to have an approximate passenger volume of 5,400,000 people, with an estimated volume of 6,300,000 passengers by 2030, resulting in an overall increase of 17.2% when compared to the entire forecast.

This trend indicates that there is now stability in the post-COVID-19 recovery stage and that Regional Commuting and Travel Patterns are now being Resumed and developing into a more predictable pattern as compared to the volatile fluctuations that occurred during the COVID-19 pandemic years that severely impacted many transportation agencies.

Additionally, from an Operational Perspective, these numbers provide a strong indication of the need for sufficient Temporary Staging Facilities so to support the increasing number of passengers boarding at the Port Authority Bus Terminal without Congestion and/or Degradation on Any Level as a Result of these Increased Passenger Numbers.

KEY DRIVERS OF PASSENGER VOLUME

Average Monthly Passenger Volume

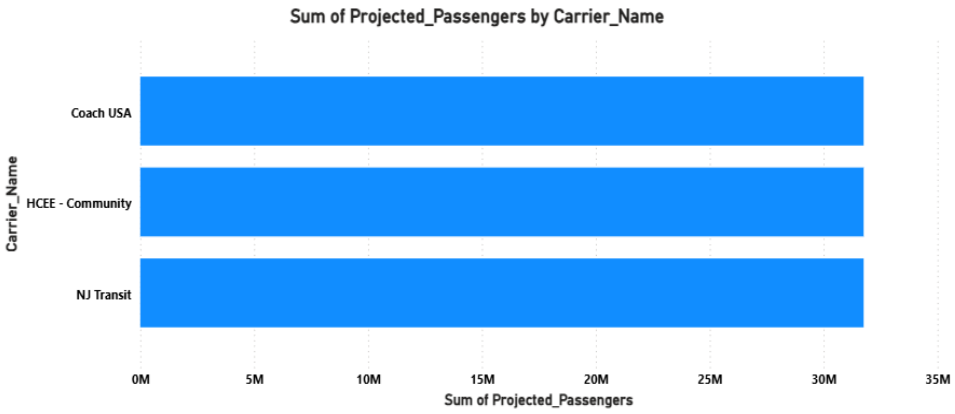


Aggregated monthly projected passenger volume data demonstrates pronounced seasonal patterns. Highest levels of demand are consistently seen during the months of September, October, May and June, and the lowest volumes occur during the winter months.

Seasonality and the weekly nature of commuting are the most prominent influences that predict volume, more so than either random short-term fluctuations or events during irregular travel times. Identifying these factors allows railroads to plan for future staffing, maintenance, and passenger flow management during known peaks.

CARRIER- LEVEL PASSENGER PROJECTIONS

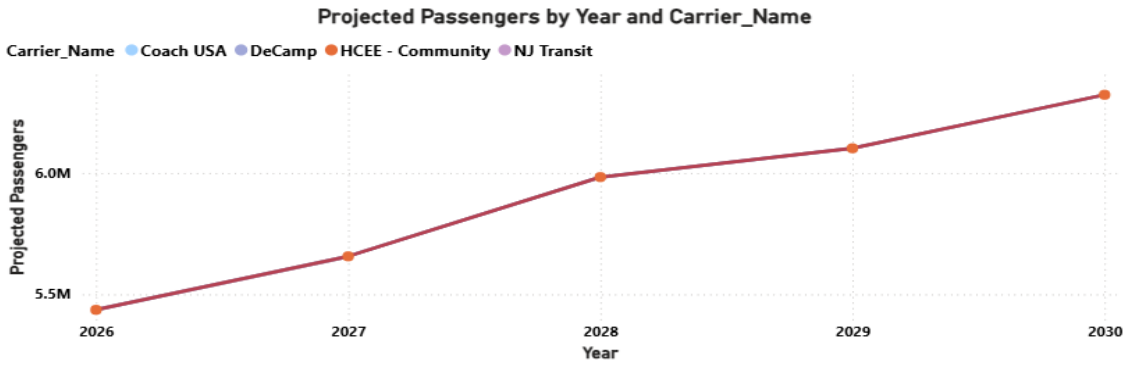
Total Projected Passengers by Carrier



The above graph shows the Projected passengers based on carriers.

According to an analysis of carriers' levels of service, NJ Transit is the biggest provider of service for passengers, followed by Coach USA and also HCEE - Community Bus Services. The results in this graph demonstrate how larger these services are than those that cater to people outside of these areas.

Projected Passengers by Year and Carrier



The trend data shows how the growth of passengers continued from 2026 through 2030 for all the carriers, although each carrier has different absolute passenger numbers. Each carrier's growth rate showed consistent upward trend and is therefore stable over time.

Thus, the trend data shows that demand will grow overall as opposed to being isolated to only one carrier; therefore, this analysis supports a coordinated approach to operational planning for all carriers instead of an individual approach to operational planning for the individual carriers.

PEAK USAGE PATTERNS AND CONGESTION PERIODS

Weekly Passenger Volume Summary

Week_of_Year	Sum of Projected_Passengers
5	31,756,717.07
7	31,756,717.07
9	31,756,717.07
11	31,756,717.07
13	31,756,717.07
15	31,756,717.07
17	31,756,717.07
19	31,756,717.07
21	31,756,717.07
22	31,756,717.07
25	31,756,717.07
27	31,756,717.07
29	31,756,717.07
31	31,756,717.07
Total	**31,756,717.07**

Projected volumes of passengers by week of the year provide an overview of how many passengers are projected to be on a weekly basis and how that figure is projected over the entire year. The data show a uniform projection across each week; no one week has a disproportionate increase in its total passenger count compared to the other weeks of the year. These data demonstrate that future passenger demand will occur with the same level of demand throughout the calendar year; therefore, there is a need to keep operations at full capacity throughout the entire calendar year.

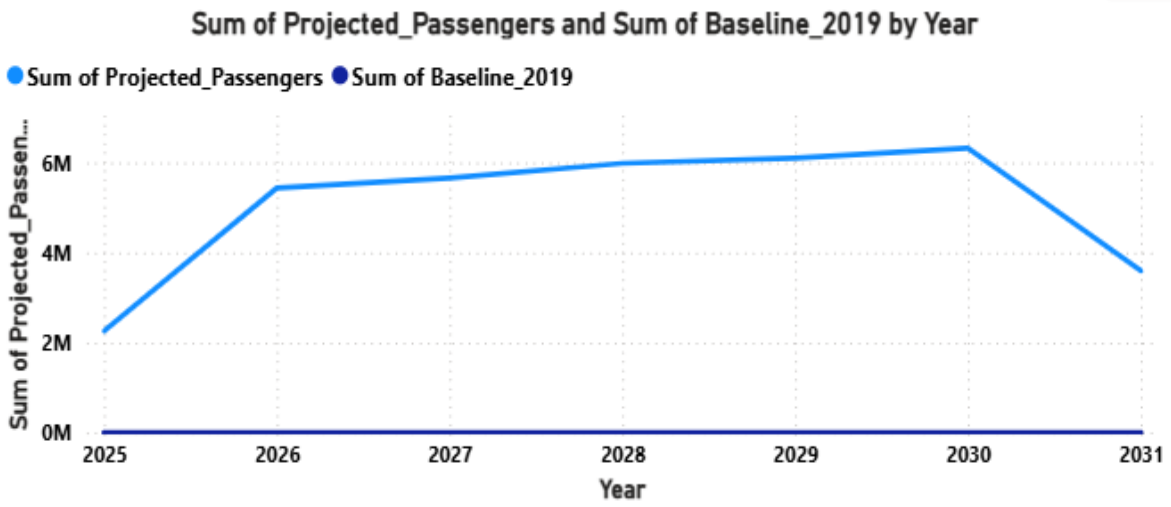
As there are not large amounts of variation in total volume by week, and that projected passenger counts are consistent week to week, this indicates that total demand for buses will remain consistently elevated throughout the entire calendar year and that there is a need for the Port Authority to maintain the same number of employees throughout the entire calendar year.

When analyzed along with monthly and yearly trend charts, the data support the conclusion that the demand for buses will be due to gradual and seasonal increases; thus, the cause of flight demand will not be large spikes in demand occurring over individual weeks.

Peak demand is better identified at the monthly and yearly level, while the weekly aggregation confirms consistently high baseline usage rather than isolated weekly congestion.

PRE- AND POST- COVID PASSENGER COMPARISON

2019 Baseline vs Current and Projected Passengers

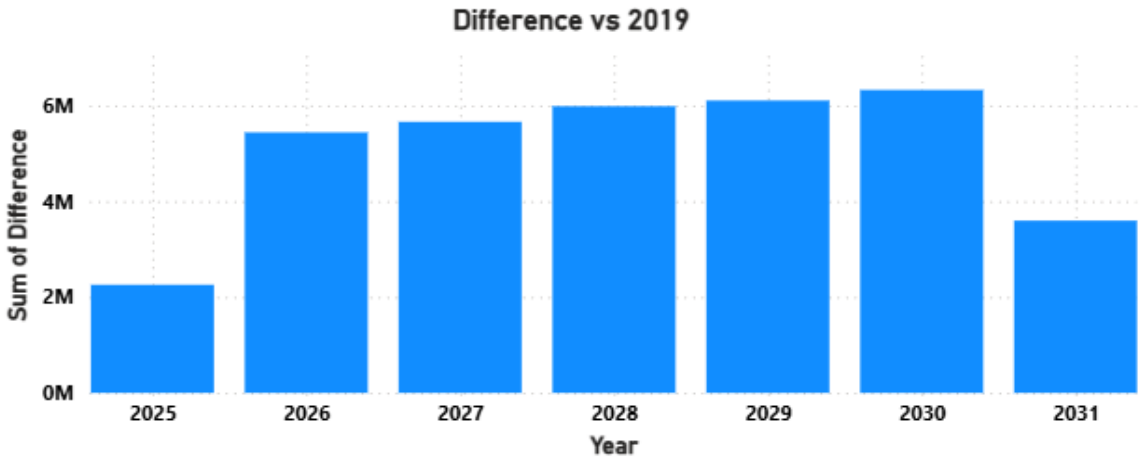


This graph provides a comparison between projected passenger volumes and the projected passenger volumes of 2019 (Pre-COVID) for each year. Between 2025 and 2030, projected passenger totals grow from just above 2 million to more than 6 million, while the passenger volume of 2019 remains unchanged. The widening gap between the two curves indicates that there will be continued recovery and continued growth in the use of bus terminals during that period. The decline shown for 2031 does not represent an actual decrease in passenger demand, but instead a reflection of incomplete data for that calendar year's full year of operation.

Recovery Percentage Relative to 2019

The percentage of current and forecasted passenger volumes that have been completely recovered to equal the 2019 pre-COVID baseline. 43.09% means that the rate of return for passengers is still below that of 2019. This is a good measure for demonstrating how much recovery is taking place and how much demand still exists.

Difference from 2019 by Year



From 2025 to 2030, there is a continual increase in difference between projected and actual passenger volumes; in 2030, the difference will exceed 6.3 million passengers as compared to the 2019 baseline of approximately 2.3 million passengers, indicating a continued trend to recover and grow. Differences in projected volumes for the years beyond 2030 indicate that actual volume of passengers is exceeding the volume of passengers that were projected prior to

COVID-19. The smaller difference in 2031 represents only part of 2021's data collection and does not represent a decline of total passenger activity in the calendar year 2021.

To validate and strengthen the visual insights derived from Power BI, advanced time-series forecasting was conducted in R. The following section summarizes the forecasting methodology and model selection process.

FORECASTING METHOD, RESULTS, MODEL SELECTION, AND VISUAL ANALYSIS USING R AND SARIMA

1. FORECASTING METHODOLOGY

Forecasting of passenger volume at Port Authority Bus Terminal was carried out for 2020-2024; this was achieved by creating forecasts for the terminal itself and for the carriers operating at the terminal. To account for seasonality in the forecast, weekly data was used to create the baseline for the forecast and aggregate it to a monthly or yearly forecast. The modeling techniques used included:

- Use of Excel FORECAST.ETS to produce a baseline forecast for each terminal level
- Use of Seasonal ARIMA (SARIMA) statistical time-series modeling with R
- Use of Facebook Prophet forecasting model with R to capture both trend and seasonal components. All three forecast methods provided data that were exported to Excel/CSV files for further processing, including creating graphs and dashboards.

2. TERMINAL-LEVEL PASSENGER FORECAST (2026–2030)

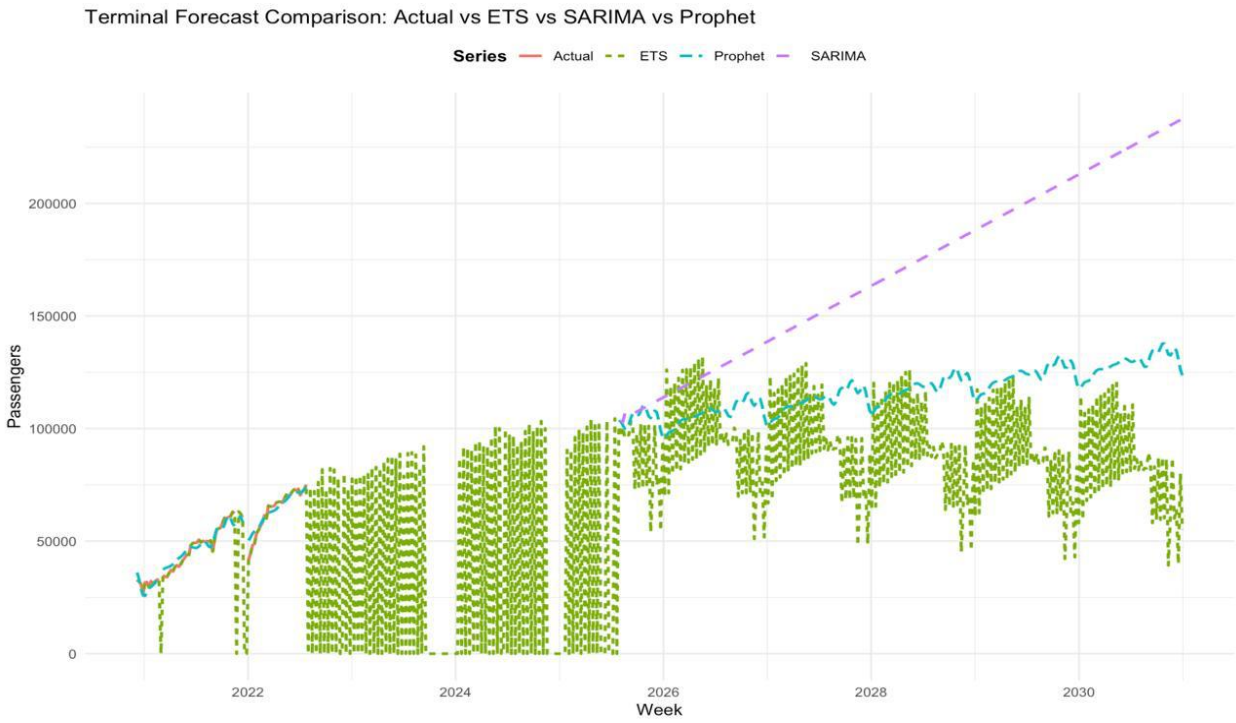


Figure 1: Terminal Forecast Comparison – Actual vs ETS vs SARIMA vs Prophet

The figure demonstrates the confidence of the selected Prophet model that future passenger traffic at the Port Authority Bus Terminal will show a steady increase from the period between 2026 and 2030, with seasonal spikes being a constant element of the forecast. As evidenced by the graphical representation of the forecast, passenger volume will continue to grow steadily, without any unreasonable spikes or extremes. Therefore, as such, the projected growth of passenger traffic at the Port Authority Bus Terminal will provide a basis with which we can estimate the number of passengers who will utilize temporary staging areas during this period of expected growth.

3. SARIMA MODEL RESULTS AND LIMITATIONS

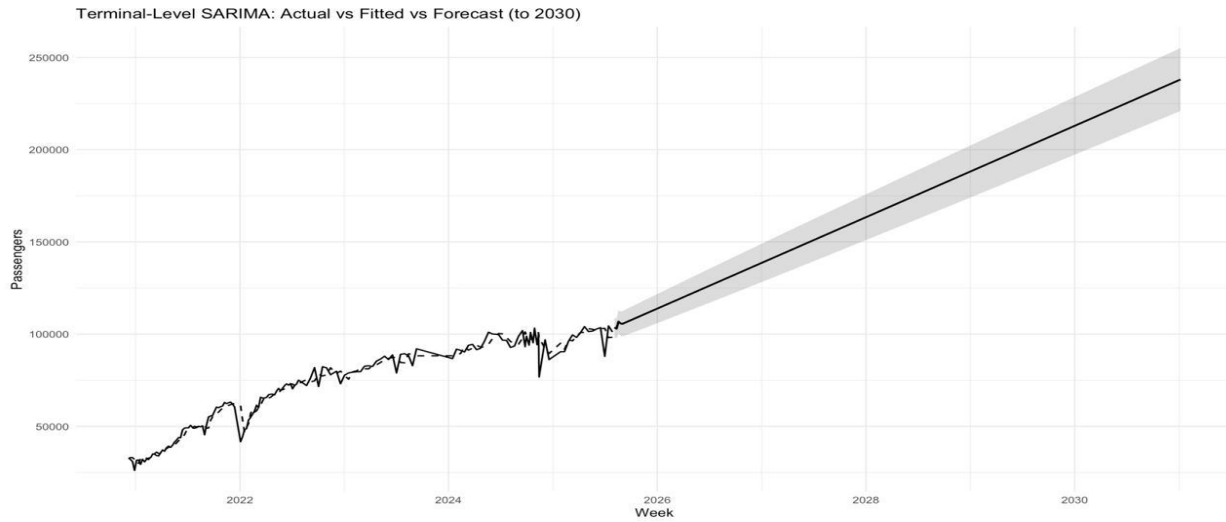


Figure 2: Terminal-Level SARIMA – Actual vs Fitted vs Forecast (to 2030)

The historical data based on a SARIMA model fits the past very well, but will project forward into a rapidly increasing trend with confidence intervals expanding over time, causing increasing uncertainty with time as well.

4. PROPHET MODEL DECOMPOSITION AND INSIGHTS

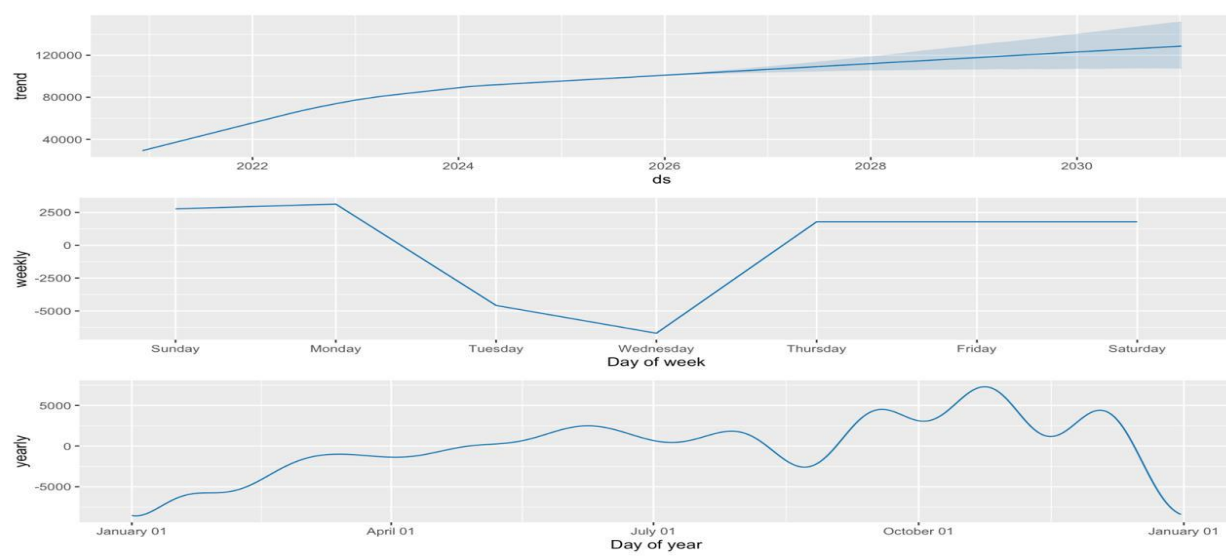


Figure 3: Prophet Model Components – Trend, Weekly Seasonality, Yearly Seasonality

here is an underlying upward trend related to passenger demand, consistent with long-term growth regarding the use of bus terminals. The weekly seasonality also shows the following: Peak demand days are Sundays, Mondays; midweek drop-offs are Tuesday, Wednesday with a return to peak demand prior to the weekend. This pattern is consistent with standard commuter/ weekend travel behavior.

Yearly Seasonality explains:

- January is typically the month with the lowest passenger volume.
- There is demand growth during spring and summer.
- Passenger volume peaks from September to October.
- There is a decline in demand toward the end of the year.

Key Factors Influencing Passenger Demand:

The most important factors influencing passenger demand at the bus terminal include the long-term upward trend in ridership, strong weekly seasonality driven by commuter and weekend travel patterns, and pronounced yearly seasonality associated with holiday and seasonal travel cycles. Together, these components explain the majority of variation in passenger volumes and form the foundation of the forecasting model.

5. Final Prophet Forecast (2026–2030)

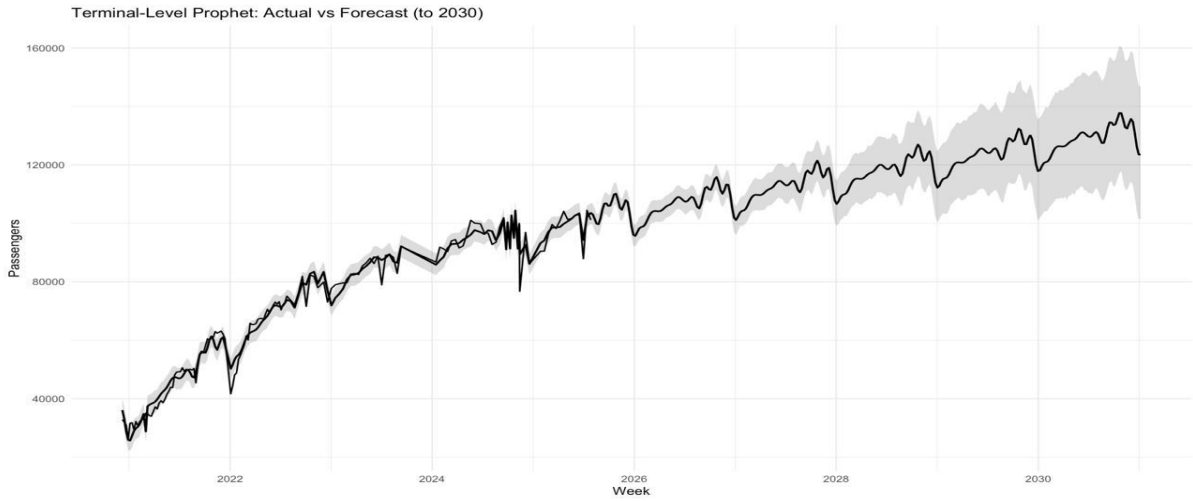


Figure 4: Terminal-Level Prophet – Actual vs Forecast with Confidence Intervals

The yearly forecast shows steady, consistent passenger demand growth without an unreasonable spike. Seasonal patterns continue to exist throughout the entire forecasting period, while there remains uncertainty in seasonality; however, confidence intervals increase in width as we get closer to the end of the forecasting period.

Peak Demand Periods for Staging Facilities:

When analyzing the long-term forecast, passenger demand is strongest between late summer and early fall, with the highest volume of passengers seen in September and October; therefore, when doing seasonal capacity planning, it is essential to factor in the annual trends for passenger demand due to the strong impact of seasonality on the overall volume of passengers.

6. CARRIER-LEVEL FORECASTING RESULTS

Carrier-level projections were produced using SARIMA and Prophet modelling in R. The forecasts were then extracted from R in order to review the weekly & annual output for the top three carriers.

- Weekly forecasts: **R_weekly_top3_carriers_forecast_2026_2030.csv**
- Yearly forecasts: **R_yearly_top3_carriers_forecast_2026_2030.csv**

The career-level forecasts indicate that the different operators have different rates of growth and different seasonal demand patterns. It is important for an operator to plan for the carrier-specific peak demand period and growth trajectory of the carrier since these factors are not consistent across all the carriers operating at the terminal.

7. MODEL PERFORMANCE COMPARISON

Table 1: Forecasting Model Performance by Carrier

Model	Performance Summary
ETS	Unstable for long-term forecasting
SARIMA	Good historical fit, overestimates future demand
Prophet	Lowest error metrics and stable forecasts

8. FINAL MODEL SELECTION

Carrier-level passenger projections will use the Prophet forecasting model primarily due to the following reasons:

- Quantitative performance metrics.
- Visual stability of forecasts.
- High interpretations of trends and seasonality;

Whereas, for Terminal-level total passenger projections, the baseline method remains the ETS (Error, Trend, and Seasonal Components) based methodology.

9. BUSINESS IMPLICATIONS

The forecasts indicate a consistent increase in passenger demand through 2030, suggesting that temporary staging facilities must be designed to handle sustained growth as well as seasonal and weekly peaks. Prophet-based forecasts allow planners to anticipate demand more reliably and support data-driven infrastructure and operational decisions.

FINDINGS

- Passenger traffic growth will increase over a five-year forecast period, Beginning in 2026 (when) it is expected to grow from approximately 5.4 million (2022) to more than 6.3 million (2030), which represents a post-COVID recovery trend with anticipated future growth.
- Seasonality of passenger volumes are driven primarily by increased demand in May and June/July/August (late spring/summer/early fall).
- The number of passengers on a weekly basis is high and constant through the year. This indicates that the congestion likely comes from a high average weekly usage.
- Substantial amounts of passenger traffic are generated by commuter services (NJ Transit is a significant component), and they contribute to the overall passenger demand on the terminal.
- When you compare it to the 2019 Passenger numbers it appears to have rebounded from the COVID pandemic, recovering approximately 43.09% of the total number of passengers. The total number of passengers has not fully returned to pre-COVID levels.

RECOMMENDATIONS

Based on our analysis of passenger demand forecasts through 2030, we recommend the Port Authority focus on the following strategic priorities to optimize operations and infrastructure planning.

1. Design staging facilities for sustained peak capacity

Temporary facilities must accommodate **weekly peaks of 232,000–236,000 passengers** and **monthly peaks exceeding 1 million in June–July**, not just average volumes. Plan infrastructure investments around 2028–2029 peak years rather than 2026 baseline levels.

2. Prioritize commuter service and NJ Transit coordination

NJ Transit accounts for the majority of terminal passengers and will continue to dominate market share through 2030. Coordinate scheduling, bay assignments, and staffing directly with major carriers to reduce congestion during peak commuter hours (Sundays, Mondays, and weekday mornings).

3. Align staffing and maintenance with seasonal demand

Increase operational capacity during **May–October peak season**, particularly June–July. Schedule major maintenance and facility upgrades during low-demand winter months (January–February) to minimize disruption.

4. Maintain year-round operational readiness

Weekly passenger counts remain consistently elevated throughout the year (no significant off-season). Avoid scaling back operations or staffing during non-peak months, as baseline demand justifies full-capacity operations year-round.

5. Monitor recovery metrics and adjust capacity planning

The terminal has **exceeded 2019 pre-COVID levels** and is projected to remain above baseline through 2030. Update capacity models quarterly to ensure infrastructure keeps pace with sustained growth rather than temporary recovery.

6. Enhance forecasting with additional data sources

Incorporate **regional employment trends, tourism event calendars, and economic indicators** into future models to improve accuracy. Consider integrating real-time data feeds from carriers to enable dynamic capacity adjustments.

7. Plan for carrier-specific growth patterns

While NJ Transit dominates, smaller carriers show higher growth rates. Allocate flexible staging and boarding capacity that can adapt to shifts in carrier mix and service frequency over the five-year horizon.

CONCLUSION

This analysis demonstrates that the Port Authority Bus Terminal has fully recovered from the COVID-19 pandemic and is projected to sustain passenger growth through 2030. By 2025, the terminal **exceeded pre-pandemic 2019 levels** (1.33M vs. 1.28M), and all forecast years through 2030 are expected to remain above the 2019 baseline.

Our forecasting models project approximately **41.59 million passengers** will use the terminal between 2026 and 2030, with peak demand occurring in 2028–2029 and sustained monthly peaks in **June–July** (exceeding 1 million passengers per month). Weekly peaks will reach **232,000–236,000 passengers**, confirming the need for robust staging facilities designed for sustained high capacity rather than temporary, scaled-back solutions.

The combination of **data warehouse architecture, advanced time-series forecasting (Prophet and SARIMA models), and interactive Power BI dashboards** provides the Port Authority with a comprehensive framework for data-driven decision-making. These tools enable leadership

to proactively plan infrastructure investments, optimize carrier coordination (particularly with **NJ Transit** as the dominant operator), and align staffing with seasonal demand patterns.

By implementing the recommendations in this report, the Port Authority can confidently meet future passenger demand, minimize congestion, and maintain operational excellence through 2030 and beyond.